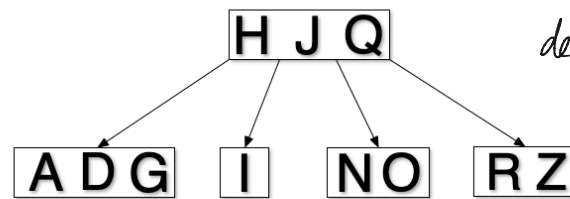

Non-Programming Part

1. The figure below shows a B-Tree with minimum degree $t=2$. The keys are sorted in alphabetical order. Using the algorithm outlined in the textbook (preemptive merge / split), please draw the final result after all of the following operations are performed in the given order: (1) insert K; (2) insert L; (3) delete K; (4) delete N; (5) delete I.



key: 1 ~ 3
degree: 2 ~ 4

